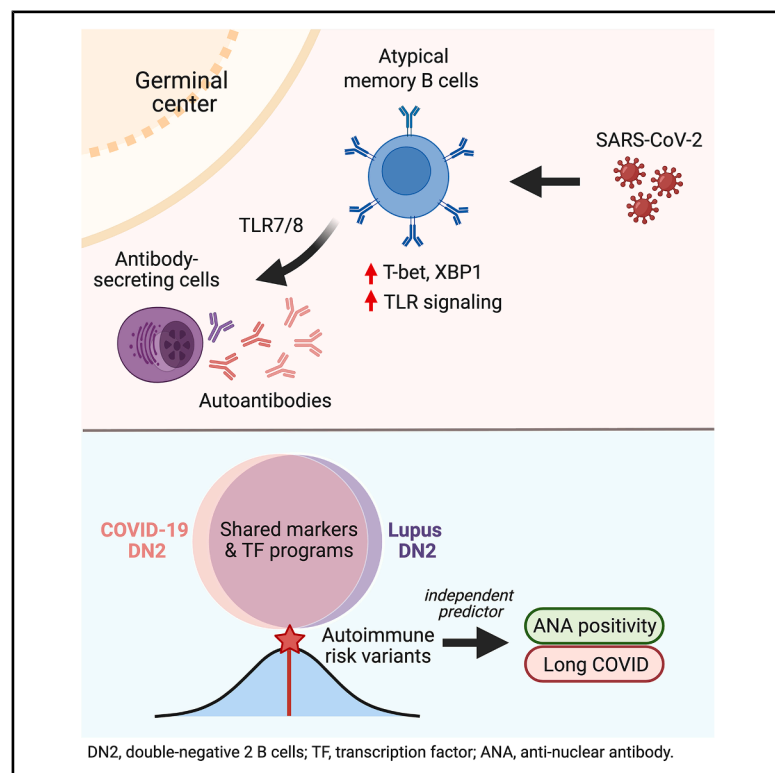


A distinct effector B cell population drives autoantibody production in SARS-CoV-2 infection

Graphical abstract



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In brief

Autoantibodies (autoAbs) are linked to mortality and Long COVID, yet their cellular origins remain unclear. Yuan et al. identify a distinct B cell population (DN2 cells) as a key source of autoAbs during SARS-CoV-2 infection. These cells mirror lupus DN2s and carry a strong autoimmune genetic signature, linking infection-triggered autoAbs to autoimmune predisposition and Long COVID.

Highlights

- AutoAb levels track atypical memory B cell (AtM) expansion and decline upon recovery
- AtMs are precursors of autoAb-producing cells in COVID-19
- DN2 B cells drive autoAb production via TLR7/8 signaling, regulated by T-bet and XBP1
- COVID-19 DN2s resemble lupus DN2s and carry the top autoimmune risk-variant signal

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